

MODULE 35 · DOMAIN 4 - ACCELERATE WORKLOAD MIGRATION AND MODERNIZATION

Server and Application Migration

Replication is the easy part. Cutover is where migrations fail.

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The service converts the **machine**. Whether the **application** works is a separate question, and you answer it by testing.

The lifecycle

01

Replicate

Agent copies blocks continuously into a staging area.

02

Test

Launch a test instance. Source keeps running, untouched.

03

Cut over

Brief pause, final sync, launch the cutover instance.

04

Finalise

Confirm, then stop replication and archive the source.

ON THE EXAM

Testing is non-disruptive and repeatable. Replication continues throughout, so you can test as many times as you like.

How replication works

- ◆ **Continuous block-level replication** into a staging subnet, using small replication servers and staging EBS volumes.
- ◆ **Outbound TCP 443 and TCP 1500** from source to AWS. No inbound access to your data centre is required.
- ◆ **Sources can be anywhere:** on-premises physical, VMware, Azure, Google Cloud, Oracle Cloud or Alibaba.
- ◆ Traffic is compressed with LZ4, typically cutting transit size by 60 to 70 per cent.

ON THE EXAM

The staging area is deliberately cheap. It is small instances and EBS volumes, not a running copy of your production fleet.

Agent-based or agentless?

AGENT-BASED

The default path

- Replication agent on each source server
- Works from any supported environment
- Physical, virtual, or another cloud
- Needs install permission on each server

AGENTLESS

VMware only

- MGN vCenter Client in the data centre
- Snapshot-based, vCenter 6.7, 7.0 and 8.0
- Same OS support as agent-based
- MGN only, not Elastic Disaster Recovery

THE TIEBREAKER

Agentless exists for estates where agents cannot be approved. Outside VMware vCenter, it is not an option at all.

Testing, and why it is skipped

- ◆ **A test instance launches from replicated data while the source keeps running.** Users notice nothing.
- ◆ Replication continues during and after the test, so you can test repeatedly as configuration changes.
- ◆ **What testing finds:** hard-coded IP addresses, hardware-bound licences, on-premises dependencies, missing firewall rules.
- ◆ Test into an isolated subnet, so a test instance cannot talk to production systems it should not.

ON THE EXAM

Every objection to testing is about time. It costs a launch and an hour, against discovering the same faults during a cutover window.

Cutover and after

- ◆ **Cutover means a brief write pause, a final sync, then launching the real instance.** Minutes, not a maintenance weekend.
- ◆ **Post-launch actions** run automatically on migrated instances: install agents, join a domain, run validation.
- ◆ Predefined actions exist, including **App2Container for replatforming** and installing observability agents.
- ◆ **Finalising archives the source and stops replication.** Do it once you are confident, not on cutover day.

ON THE EXAM

Post-launch actions are how you avoid 800 servers each needing a human to install the same three agents afterwards.

Cost, and a naming note

- ◆ **Free for 2,160 hours per source server**, which is 90 days of continuous replication, starting when the agent is installed.
- ◆ **Staging infrastructure is charged throughout**, including during the free period. The service is free; the EC2 and EBS are not.
- ◆ After 90 days, roughly \$0.042 per server-hour. A migration that drifts across many servers adds up.
- ◆ **Naming:** the service has been rebranded **AWS Transform MGN**. The exam's list still says Application Migration Service.

ON THE EXAM

The free period starts at agent install, not at cutover. Installing agents early on servers you will migrate last wastes it.

Reading the scenario

THE SCENARIO SAYS

Migrate servers from another cloud

No agents may be installed, VMware estate

Cutover failed and nobody knew why

Install agents on every migrated server

Convert Oracle to PostgreSQL

Replication has run for six months

IT IS POINTING AT

MGN. Sources can be anywhere

Agentless replication, vCenter Client

Test instances were never launched

Post-launch actions

DMS and SCT, not MGN

The 90-day free period has expired

ON THE EXAM

MGN moves a machine as it is. Anything that changes the engine, the schema or the architecture is a different tool.

Meridian Freight must move 300 servers from another public cloud provider to AWS within four months, following a commercial dispute. The servers run a mix of Windows and Linux, and the applications must not be modified.

The team assumes AWS migration tooling only works from on-premises data centres.

What should the solutions architect recommend?

- A Application Migration Service, installing the replication agent on the source servers in the other cloud
- B Export each server as an image, transfer it to S3, and import with VM Import/Export
- C Rebuild all 300 servers natively in AWS
- D Use AWS DataSync to copy the servers

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

MGN from the other cloud

MGN works anywhere the agent can reach AWS on TCP 443 and 1500, including Azure, Google Cloud and others. The team's assumption is simply wrong.
- B

Export and VM Import

A point-in-time copy, so the source keeps changing while you transfer. It also means a long outage per server at cutover.
- C

Rebuild natively

The applications must not be modified, and 300 rebuilds in four months is not achievable regardless.
- D

DataSync

Moves file and object data. It does not migrate servers, operating systems or applications.

THE SIGNAL

The team's stated assumption is the question. MGN is a cloud-to-cloud migration tool as much as an on-premises one.

A financial services firm must migrate 450 VMware virtual machines running on vCenter 8. Its change control board will not approve installing any agent on production virtual machines, and the exception process takes six months.

The migration must begin within four weeks.

What should the solutions architect recommend?

- ☐ A Deploy the MGN vCenter Client and use agentless replication
- ☐ B Begin the exception process for agent installation
- ☐ C Use Elastic Disaster Recovery in agentless mode instead
- ☐ D Export the VMs as OVA files and import them individually

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A** Agentless via the vCenter Client
Exactly the case agentless exists for. One client in the data centre, nothing on the VMs, and vCenter 8 is supported.
- B** Exception process
Six months against a four-week start. It also spends political capital to obtain something agentless replication already provides.
- C** DRS agentless
Agentless replication is available in MGN only. Elastic Disaster Recovery does not offer it, and DRS is for disaster recovery anyway.
- D** Export OVA files
450 point-in-time exports, manually imported, with the source drifting throughout. It does not scale and does not sync.

THE SIGNAL

The vCenter version matters. Agentless supports 6.7, 7.0 and 8.0, so vCenter 8 is deliberately in scope here.

A cutover of 40 servers over a weekend failed. Twelve applications would not start, citing licence errors and unreachable database hosts. The team rolled back after nine hours. Replication had reported healthy for weeks, and no test instances were ever launched. The next cutover window is in three weeks.

What should the solutions architect recommend before the next attempt?

- ☐ A Launch test instances into an isolated subnet, resolve the licence and connectivity failures, and repeat until clean
- ☐ B Extend the cutover window to a full weekend so there is time to fix problems
- ☐ C Re-install the replication agents and restart replication from scratch
- ☐ D Migrate the 12 failing applications last, and cut over the other 28 servers

◆ PAUSE AND DECIDE

What should the solutions architect recommend before the next attempt?

- A

Test instances, iterate until clean
Non-disruptive and repeatable, with the source still serving users. Licence binding and unreachable hosts are exactly what a test finds.
- B

Longer window
Turns a nine-hour failure into a longer one. Debugging licence activation under cutover pressure is the worst possible time.
- C

Restart replication
Replication was healthy and is not the fault. Restarting discards weeks of transferred data for no benefit.
- D

Defer the twelve
Cuts over 28 servers that were also never tested. The same class of fault is waiting in them.

THE SIGNAL

Replication healthy for weeks is the planted detail. It tells you the copying worked and the application never did.

A migration includes an Oracle database that must become Amazon Aurora PostgreSQL, and 60 application servers that must move unchanged. A junior architect proposes using Application Migration Service for all of it, reasoning that MGN replicates whole disks including the database files.

The application code will be updated separately.

What should the solutions architect advise?

- ☐ A MGN for the 60 application servers, and DMS with the Schema Conversion Tool for the Oracle to Aurora PostgreSQL move
- ☐ B MGN for everything, as proposed
- ☐ C DMS for everything, including the application servers
- ☐ D Rebuild the application servers and use MGN only for the database

◆ PAUSE AND DECIDE

What should the solutions architect advise?

- A** MGN for servers, DMS and SCT for the engine change

MGN moves machines as they are. Oracle to Aurora PostgreSQL is an engine conversion, which needs schema conversion and data migration.
- B** MGN for everything

MGN would faithfully replicate an Oracle database onto EC2 running Oracle. The requirement was Aurora PostgreSQL, which it cannot produce.
- C** DMS for everything

DMS migrates databases. It does not migrate application servers or operating systems.
- D** Rebuild the servers

The servers must move unchanged, and rebuilding 60 of them is work the requirement explicitly avoids.

THE SIGNAL

The junior architect's reasoning is correct and irrelevant. MGN does replicate database files, faithfully, onto the same engine.

After migrating 200 servers, the operations team must install the CloudWatch agent, the Systems Manager agent and a security agent on every instance, and join each to Active Directory. Doing this manually took three days for the first wave of 40. Six more waves are planned.

What should the solutions architect recommend?

- ☐ A Configure post-launch actions in MGN so each migrated instance is configured automatically at launch
- ☐ B Write a Run Command document and execute it after each wave
- ☐ C Build a golden AMI containing the agents and migrate onto it
- ☐ D Accept the three days per wave as unavoidable

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

Post-launch actions
Built for this. Each migrated instance is configured automatically as it launches, including test launches, so problems surface during testing.
- B

Run Command per wave
The strong near miss. It automates the work and still needs a human to trigger it, after launch, per wave.
- C

Golden AMI
MGN replicates the source disk. A migrated instance boots from replicated data, not from an AMI you prepared.
- D

Accept it
Three days times seven waves is three working weeks of manual configuration, with the errors that implies.

THE SIGNAL

Post-launch actions also run on test launches, so configuration failures appear during testing rather than at cutover.

A team installed replication agents on all 600 servers in month one to "get replication out of the way". Migration waves run monthly, and the final wave completes in month seven. Finance queries an unexpected Application Migration Service charge appearing from month four.

The team believed the service was free.

What should the solutions architect explain, and recommend?

- ☐ A The 90-day free period starts at agent installation; install agents per wave, shortly before each wave migrates
- ☐ B The charge is for staging EC2 and EBS, which cannot be avoided
- ☐ C The free period covers the whole account, not per server
- ☐ D Request a service credit, since the service is advertised as free

◆ PAUSE AND DECIDE

What should the solutions architect explain, and recommend?

- A** Free period starts at agent install
2,160 hours per server from install, so agents fitted in month one expire in month four. Install per wave and most servers stay free.
- B** It is staging infrastructure
Staging EC2 and EBS are charged, and separately and from day one. A charge appearing in month four is the free period expiring.
- C** Account-wide free period
The free period is per source server, not per account. That is what makes staggered installation worthwhile.
- D** Request a credit
The terms were met. The service was free for 90 days per server, and the team consumed that on servers it was not yet migrating.

THE SIGNAL

Month four is the arithmetic. Ninety days from a month-one install lands exactly there, which is the clue the scenario gives you.

What to carry out of this module

- ◆ **Replication converts the machine; testing proves the application.** They are separate questions.
- ◆ **Test instances are non-disruptive and repeatable.** The source keeps serving users throughout.
- ◆ **Agentless is VMware only, and MGN only.** Elastic Disaster Recovery does not offer it.
- ◆ **The free period starts at agent install,** not at cutover. Install per wave, not all at once.

ON THE EXAM

MGN moves a machine as it is. An engine conversion is DMS and SCT; MGN handles the application tier alongside it.